Topic 2 – Cells and control

Students should:		Maths skills
2.1	Describe mitosis as part of the cell cycle, including the stages interphase, prophase, metaphase, anaphase and telophase and cytokinesis	
2.2	Describe the importance of mitosis in growth, repair and asexual reproduction	
2.3	Describe the division of a cell by mitosis as the production of two daughter cells, each with identical sets of chromosomes in the nucleus to the parent cell, and that this results in the formation of two genetically identical diploid body cells	
2.4	Describe cancer as the result of changes in cells that lead to uncontrolled cell division	
2.5	Describe growth in organisms, including: a cell division and differentiation in animals b cell division, elongation and differentiation in plants	
2.6	Explain the importance of cell differentiation in the development of specialised cells	
2.7	Demonstrate an understanding of the use of percentiles charts to monitor growth	1c 4a
2.8	Describe the function of embryonic stem cells, stem cells in animals and meristems in plants	1d
2.9	Discuss the potential benefits and risks associated with the use of stem cells in medicine	
2.10B	Describe the structures and functions of the brain including the cerebellum, cerebral hemispheres and medulla oblongata	
2.11B	**Explain how the difficulties of accessing brain tissue inside the skull can be overcome by using CT scanning and PET scanning to investigate brain function**	1d 2d
2.12B	**Explain some of the limitations in treating damage and disease in the brain and other parts of the nervous system, including spinal injuries and brain tumours**	
2.13	Explain the structure and function of sensory receptors, sensory neurones, relay neurones in the CNS, motor neurones and synapses in the transmission of electrical impulses, including the axon, dendron, myelin sheath and the role of neurotransmitters	2g 4a, 4c
2.14	Explain the structure and function of a reflex arc including sensory, relay and motor neurones	

Students should:		Maths skills
2.15B	Explain the structure and function of the eye as a sensory receptor including the role of: a the cornea and lens b the iris c rod and cone cells in the retina	2c
2.16B	Describe defects of the eye including cataracts, long-sightedness, short-sightedness and colour blindness	
2.17B	Explain how cataracts, long-sightedness and short-sightedness can be corrected	

Use of mathematics

- Use estimations and explain when they should be used (1d).
- Use percentiles and calculate percentage gain and loss of mass (1c).
- Translate information between numerical and graphical forms (4a).
- Use a scatter diagram to identify a correlation between two variables (2g).
- Extract and interpret information from graphs, charts and tables (2c and 4a).
- Extract and interpret data from graphs, charts, and tables (2c).
- Understand and use percentiles (1c).
- Use fractions and percentages (1c).

Suggested practicals

- Investigate human responses to external stimuli.
- Investigate reaction times.
- Investigate the speed of transmission of electrical impulses in the nervous system.